

ATARNotes

Specialist Maths 1/2

July 2020 – Livestream Lecture

By Daniel

- Hi, I'm Daniel
- I graduated in 2017 with 99.90 ATAR
- This is a photo of me when I was doing Spesh!
- I'm currently studying Data Science at ~~Hogwarts~~ Unimelb



HOW YOU CAN BENEFIT FROM TODAY'S LECTURE

- Ask me any question via the poll section by just scrolling down in your browser



- Please get a pen & papers to brainstorm questions
- For practice questions, there'll be 60 seconds for brainstorm – so don't go away!

Overview

IN TODAY'S LECTURE

- Trigonometry
 - Break 1
- Coordinate Systems
 - Break 2
- Vectors

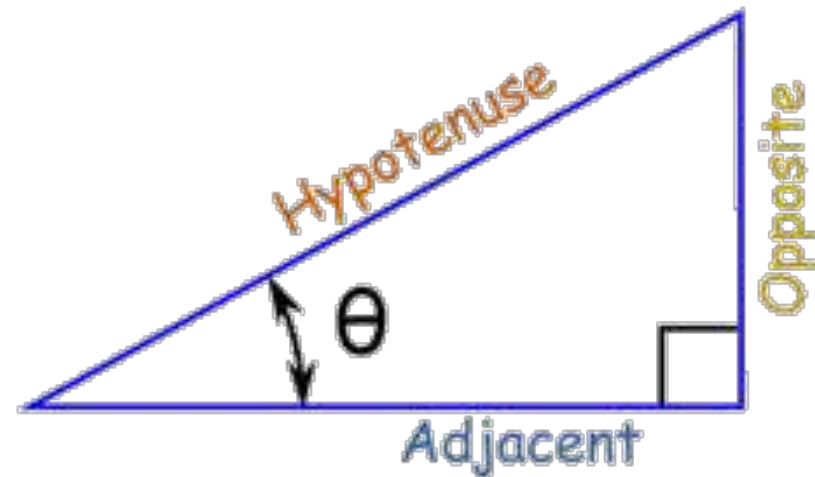


BASIC DEFINITIONS

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

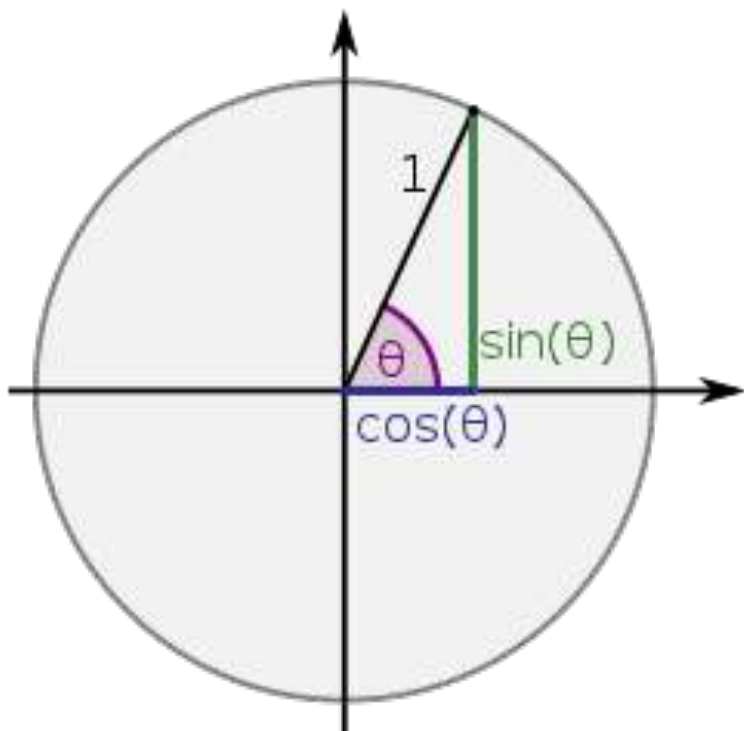
$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$



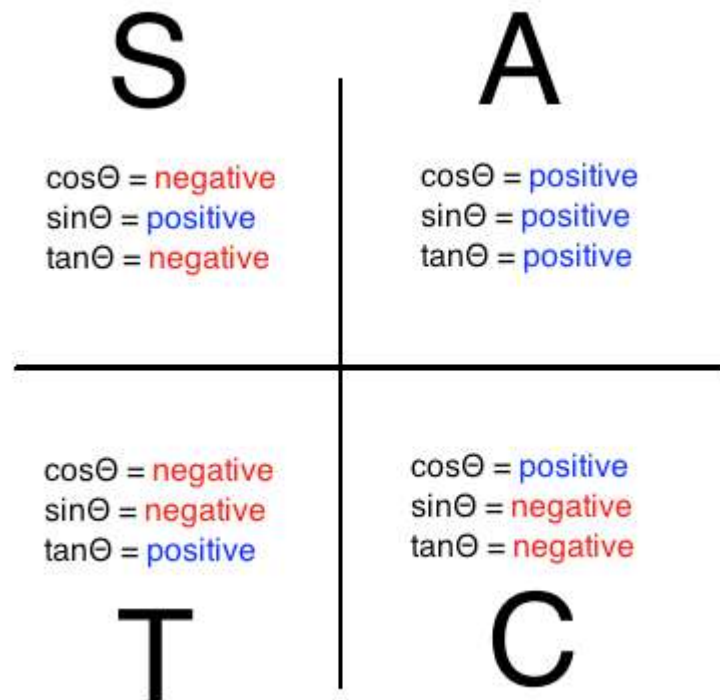
UNIT CIRCLE DEFINITION

A note about θ



- Positive θ are measured *anti-clockwise*
- Negative θ are measured *clockwise*
- Even though θ is typically between 0 and 360 degrees, there is strictly no limit for θ .
- $\theta > 360$ means some revolutions!

UNIT CIRCLE DEFINITION



A note about θ

- Positive θ are measured *anti-clockwise*
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- Even though θ is typically between 0 and 360 degrees, there is strictly no limit for θ .
- $\theta > 360$ means some revolutions!

EXACT VALUES

One thing that you need to know by heart (for both Methods and Spesh) is the exact values of basic angles. Here's a trick for how to do it

	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin(x)$					
$\cos(x)$					
$\tan(x)$					

SYMMETRICAL VALUES

- $\cos(-x) = \cos(x)$
- $\sin(\pi - x) = \sin(x)$
- $\tan(\pi + x) = \tan(x)$

- $\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$
- $\cos\left(\frac{\pi}{2} + x\right) = -\sin(x)$

- $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$
- $\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$

FIND EXACT VALUES

Evaluate $\sin(855)$

$$\sin(855)$$

$$= \sin(360 \times 2 + 135)$$

$$= \sin(135)$$

$$= \sin(45 + 90)$$

$$= -\sin(45)$$

$$= -\frac{\sqrt{2}}{2}$$

Evaluate $\tan\left(-\frac{5\pi}{6}\right)$

$$\tan\left(-\frac{5\pi}{6}\right)$$

$$= \tan\left(\frac{\pi}{6} - \pi\right)$$

$$= \tan\left(-\left(\pi - \frac{\pi}{6}\right)\right)$$

$$= -\tan\left(\pi - \frac{\pi}{6}\right)$$

$$= -\left(-\tan\left(\frac{\pi}{6}\right)\right)$$

$$= \tan\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{3}$$

COMPOUND & DOUBLE ANGLES

- $\sin(u \pm v) = \sin(u)\cos(v) \pm \sin(v)\cos(u)$
- $\cos(u \pm v) = \cos(u)\cos(v) \mp \sin(u)\sin(v)$
- $\sin(2x) = 2\sin(x)\cos(x)$
- $\cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1 = 1 - 2\sin^2(x)$

FIND EXACT VALUES

Evaluate $\cos(75)$

$$\cos(75)$$

$$= \cos(30 + 45)$$

$$= \cos(30) \cos(45) - \sin(30) \sin(45)$$

$$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} - \frac{1}{2} \times \frac{\sqrt{2}}{2}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

Evaluate $\sin(5\pi/24)$

$$\cos(75) = \cos\left(\frac{5\pi}{12}\right) = \cos\left(2 \times \frac{5\pi}{24}\right)$$

$$= 1 - 2\sin^2\left(\frac{5\pi}{24}\right) = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\Rightarrow \sin^2\left(\frac{5\pi}{24}\right) = \frac{1}{2} \left(1 - \frac{\sqrt{6} - \sqrt{2}}{4}\right)$$

$$= \frac{4 - \sqrt{6} + \sqrt{2}}{8}$$

$$\Rightarrow \sin\left(\frac{5\pi}{24}\right) = \sqrt{\frac{4 - \sqrt{6} + \sqrt{2}}{8}} \approx 0.6088$$

TRIGONOMETRIC IDENTITIES

Identities are equations that always hold regardless of the angle.

$$\sin^2(x) + \cos^2(x) = 1$$

$$\frac{\sin(x)}{\cos(x)} = \tan(x)$$

$$\sec^2(x) - \tan^2(x) = 1$$

$$\operatorname{cosec}^2(x) - \cot^2(x) = 1$$

$$\cot(x) = \frac{1}{\tan(x)}$$

FIND EXACT VALUES

$$\sec(x) = -\frac{3}{2} \mid \frac{\pi}{2} \leq x \leq \pi. \text{ Find } \sin(x)$$

- $\sec(x) = -\frac{3}{2} \Rightarrow \cos(x) = -\frac{2}{3}$
- $\cos^2(x) + \sin^2(x) = 1 \Rightarrow \left(-\frac{2}{3}\right)^2 + \sin^2(x) = 1$
- $\Rightarrow \sin(x) = \pm \frac{\sqrt{5}}{3}$
- $\frac{\pi}{2} \leq x \leq \pi \Rightarrow \sin(x) > 0 \Rightarrow \sin(x) = \frac{\sqrt{5}}{3}$

Question 5 (5 marks)

- a. For the function with rule $f(x) = 96 \cos(3x) \sin(3x)$, find the value of a such that $f(x) = a \sin(6x)$.

1 mark



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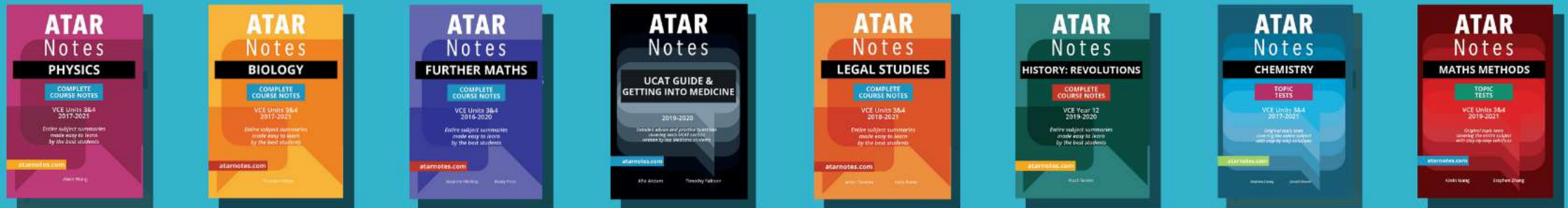


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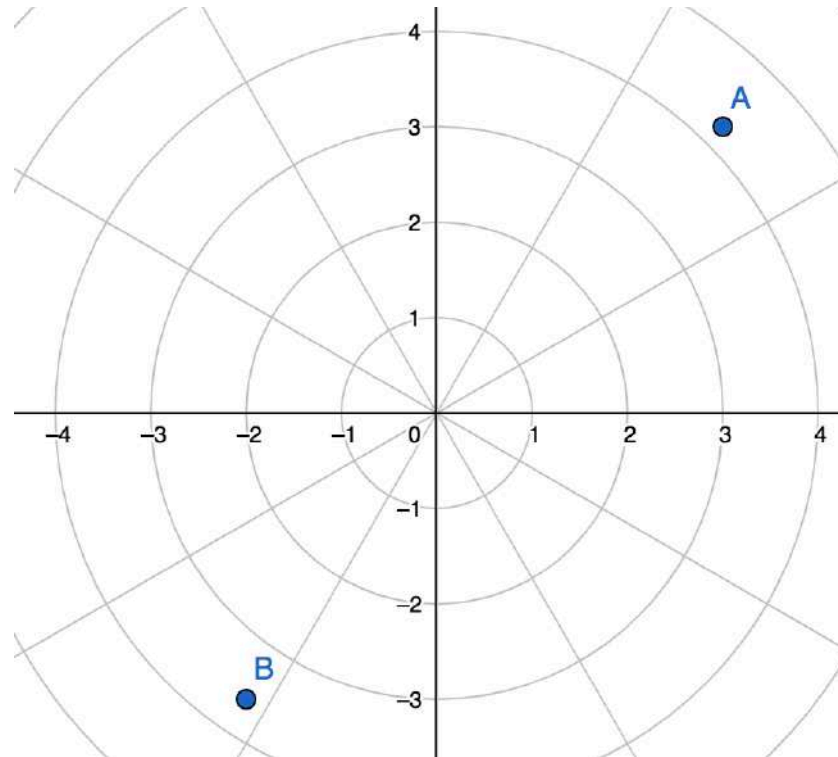
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- How do we represent the position of a **static** point on a 2D plane ?



- How do we represent the position of a **static** point on a plane ?
- Using either:
 - A pair of horizontal & vertical ordinates $(x, y) \rightarrow$ Cartesian
 - A pair of distance & angle $(r, \theta) \rightarrow$ **Polar**
- Normally, we use Cartesian system when dealing with functions
- We use Polar system when dealing with rotational quantities (vectors and complex numbers)

CONVERTING POLAR – CARTESIAN

- Polar to Cartesian:

$$x = r \cos(\theta) , y = r \sin(\theta)$$

- Use trig identities if you need to!

- Cartesian to Polar:

$$r = \sqrt{x^2 + y^2} , \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Convert the function $r = \tan(\theta) \sec(-\theta)$ into Cartesian form

- $y = r \sin(\theta) = \tan(\theta) \sec(-\theta) \sin(\theta)$

$$= \frac{\sin(\theta)}{\cos(-\theta)} \cdot \frac{1}{\cos(\theta)} \cdot \sin(\theta) = \tan^2(\theta)$$

- $x = r \cos(\theta) = \tan(\theta) \sec(-\theta) \cos(\theta)$

$$= \frac{\sin(\theta)}{\cos(\theta)} \cdot \frac{1}{\cos(-\theta)} \cdot \cos(\theta) = \tan(\theta)$$

- $y = x^2$

IF YOU THINK CARTESIAN GRAPHING IS BAD...

<https://www.desmos.com/calculator/ocpwsg5z8e>



- How do we represent the path of a **moving** point on a plane ?
- With Cartesian system, the path is represented by a function.
- But what if we're required to find the exact position at any time t ?
- Idea: instead of using (x, y) we use $(f(t), g(t))$
- t is called the parametric variable

CONVERTING PARAMETRIC – CARTESIAN

- Parametric to Cartesian:
 - To eliminate t from both equations $x = f(t)$, $y = g(t)$
 - Find $t = f^{-1}(x)$
 - Substitute into $y = g[f^{-1}(x)]$
 - Sometimes helpful to use trigonometric identities
- Cartesian to Parametric: By recognition

CONVERTING PARAMETRIC – CARTESIAN

A point moves according to the parametric equations:

$$x = t - 2$$

$$y = t^2 - 4t + 1$$

Show that the Cartesian equation of the path is $y = x^2 - 3$

$$x = t - 2 \Rightarrow t = x + 2$$

$$y = t^2 - 4t + 1$$

$$= (x + 2)^2 - 4(x + 2) + 1$$

$$= x^2 - 3$$

$$\Rightarrow y = x^2 - 3$$

2014 SPECMATH EXAM 1

4

Question 2 (5 marks)

The position vector of a particle at time $t \geq 0$ is given by

$$\underline{r}(t) = (t - 2)\underline{i} + (t^2 - 4t + 1)\underline{j}$$

- a. Show that the cartesian equation of the path followed by the particle is $y = x^2 - 3$.



1 mark

MOVING PATH INTERSECTION

Two points $A(2t - 10, 3)$ and $B(2, t + 4)$ starts moving at $t = 0$.

When will their paths intersect?

- *When A intersects B:*

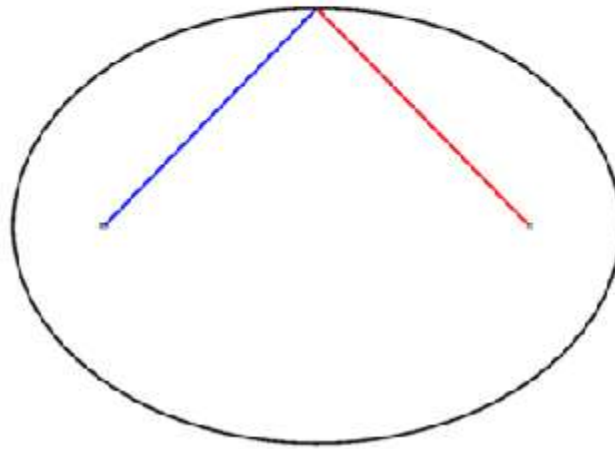
$$\begin{array}{lcl} x_A = x_B & \Rightarrow & 2t - 10 = 2 \Rightarrow t = 6 \\ y_A = y_B & \Rightarrow & 3 = t + 4 \Rightarrow t = -1 \end{array}$$

- *There is no t such that $(x_A, y_A) = (x_B, y_B)$; so the paths will never intersect*

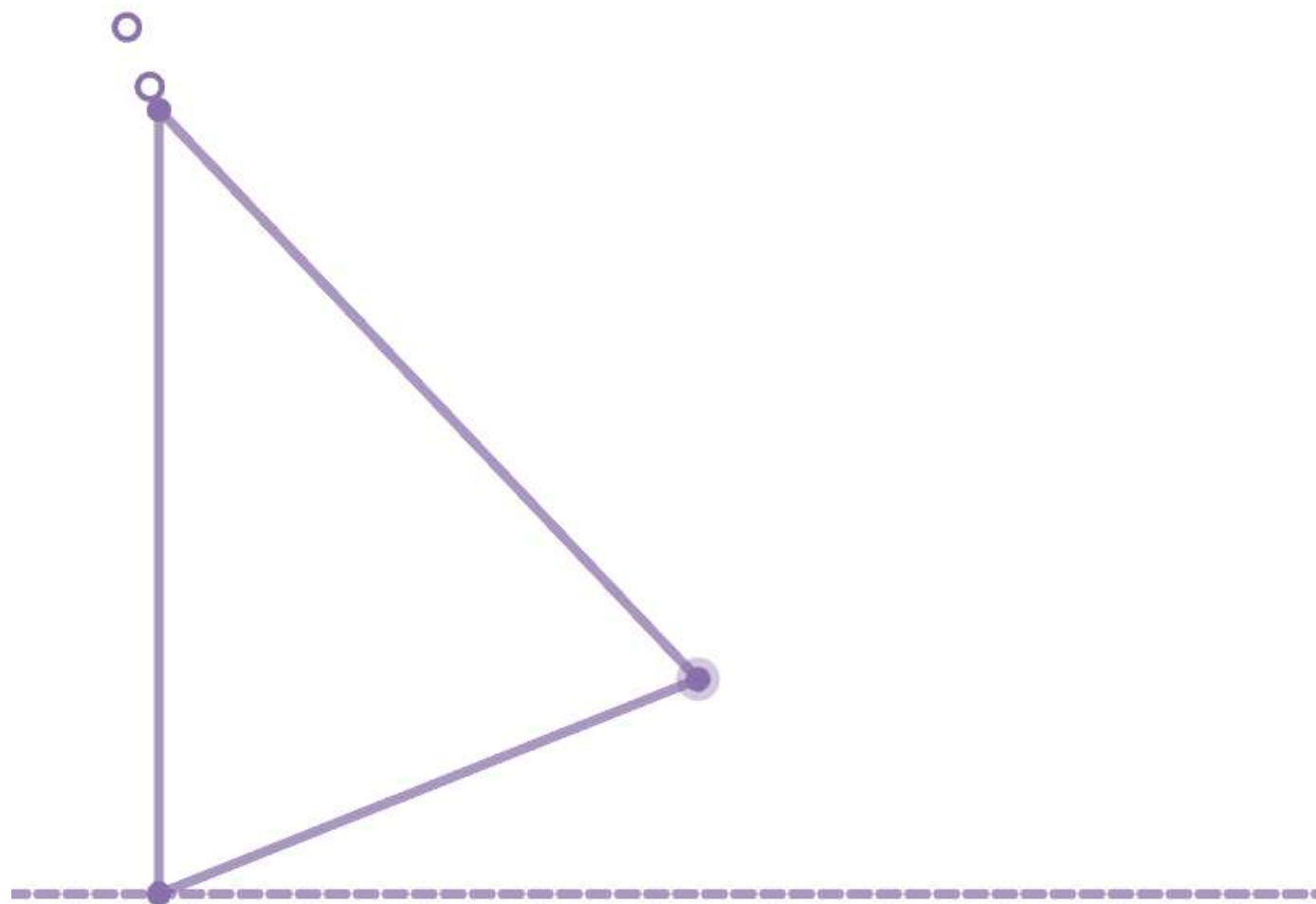
- What if the motion is defined by a specific rule?
- E.g. a point that moves in a plane so that its distance from the origin is always 3 → This is a circle, centre at origin, radius 3
- The path traced by the motion of a point following a rule is called **the locus of the point**.
- There can be different loci for the same shape.
- Usually the rules are distance-based and regarding some fixed points

MOVING WITH A RULE

Blue line length: _____ (5.000)
Red line length: _____ (5.000)
Total length: _____ (10.00)



MOVING WITH A RULE



CONVERTING LOCUS TO CARTESIAN

Find the Cartesian equation of the line that is equidistant from the point $(1, \sqrt{3})$ and the origin.

Let the line be the locus of the point $P(x, y)$

Since P is equidistant from $(1, \sqrt{3})$ and $(0,0)$:

$$\sqrt{(x-1)^2 + (y-\sqrt{3})^2} = \sqrt{(x-0)^2 + (y-0)^2}$$

$$(x-1)^2 + (y-\sqrt{3})^2 = x^2 + y^2$$

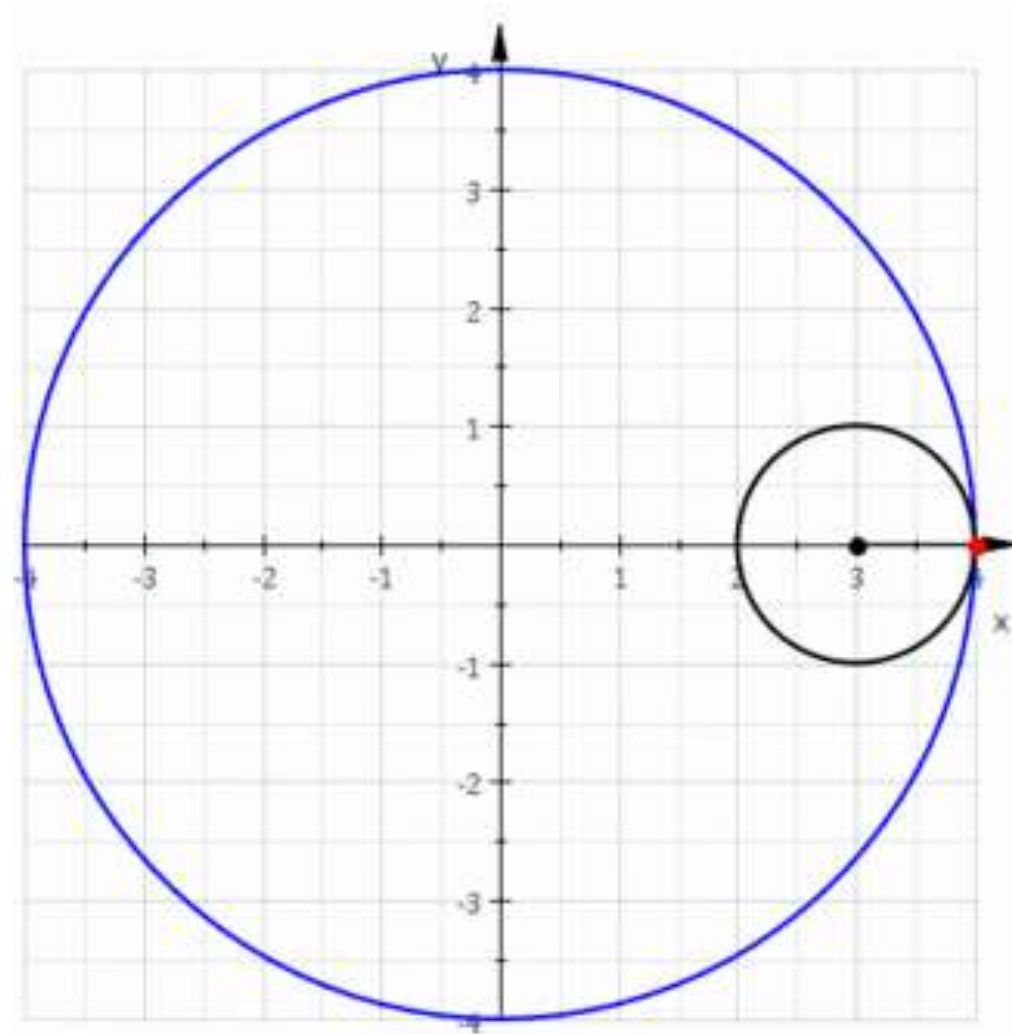
iii. Use the fact that the line $|z - (1 + i\sqrt{3})| = |z|$ passes through the point $z = 2$, or otherwise, to find the equation of this line in cartesian form.

1 mark

$$x^2 - 2x + 1 + y^2 - 2\sqrt{3}y + 3 = x^2 + y^2$$

$$-2x - 2\sqrt{3}y + 3 = 0 \Rightarrow y = -\frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{2}$$

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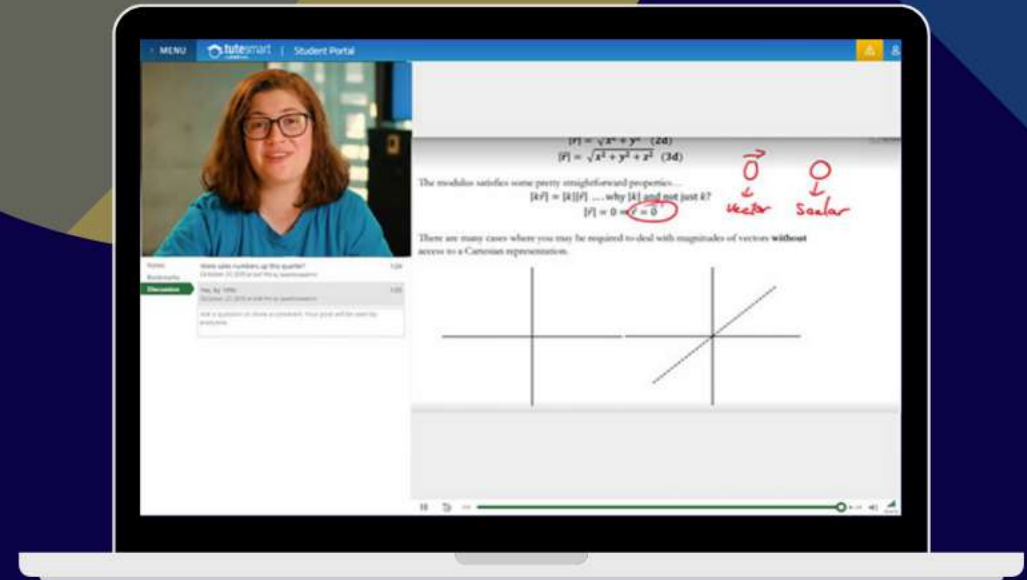
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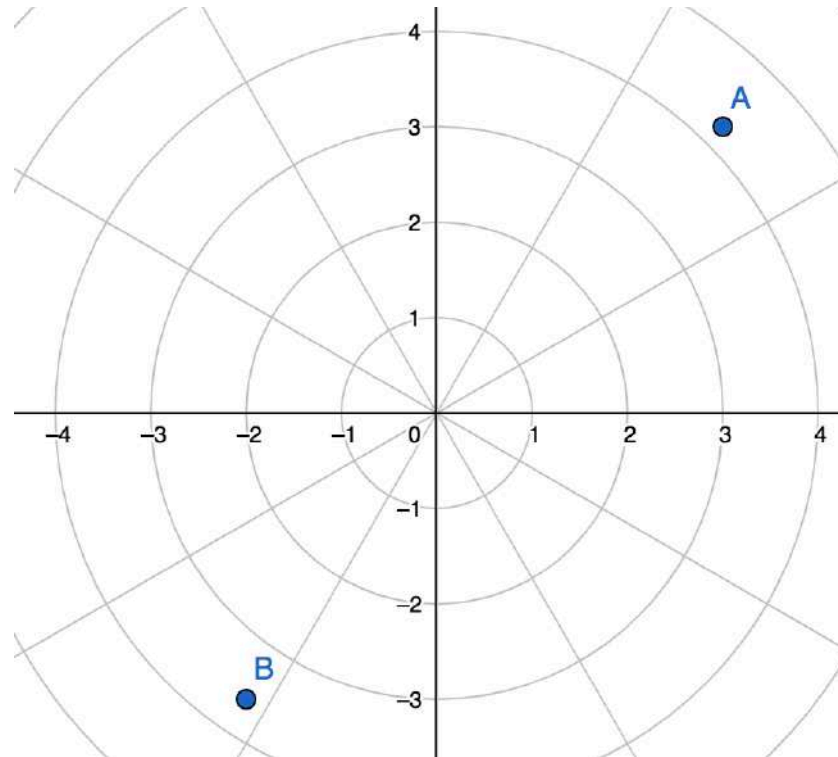
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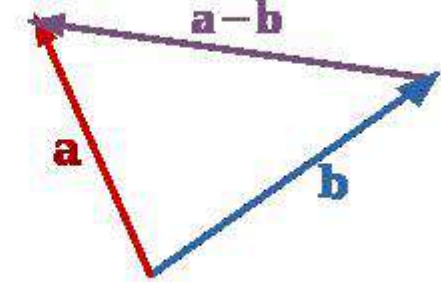
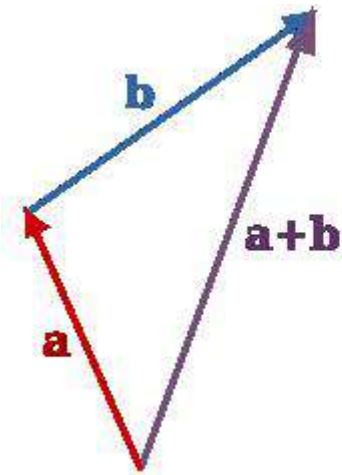
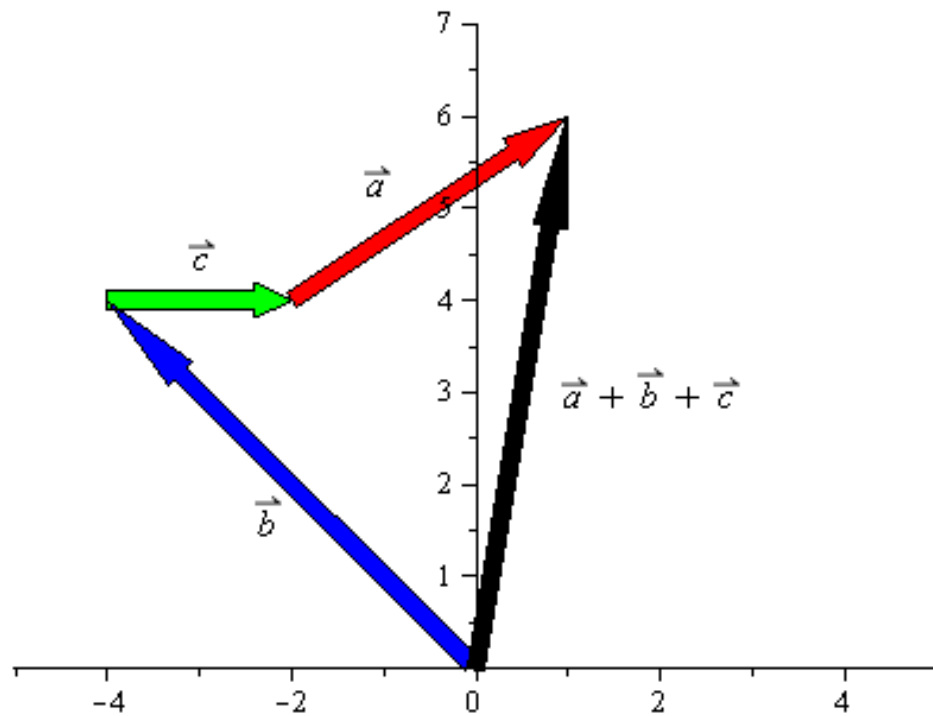
UNPOPULAR DEFINITION

- Vectors represents the fixed straight motions of a point from origin.



- $\vec{u} = a\vec{i} + b\vec{j} = \begin{bmatrix} a \\ b \end{bmatrix}$
- If $A(x_A, y_A)$ and $B(x_B, y_B)$ then $\overrightarrow{AB} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j}$
- Magnitude (length) of a vector: $|\vec{u}| = \sqrt{a^2 + b^2}$

- $|\vec{a}| + |\vec{b}| \neq \vec{a} + \vec{b}$



- This is a special operation for vectors. There's no geometric representation of the dot product!
- Vector “times” Vector = a Number!

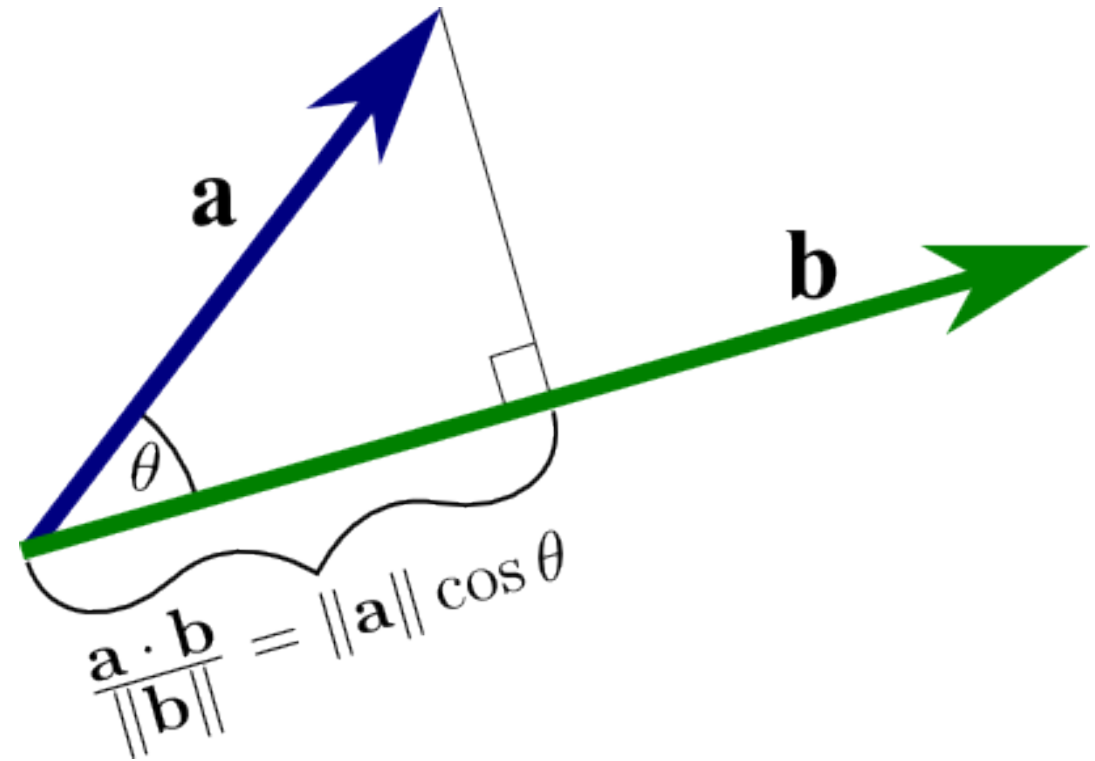
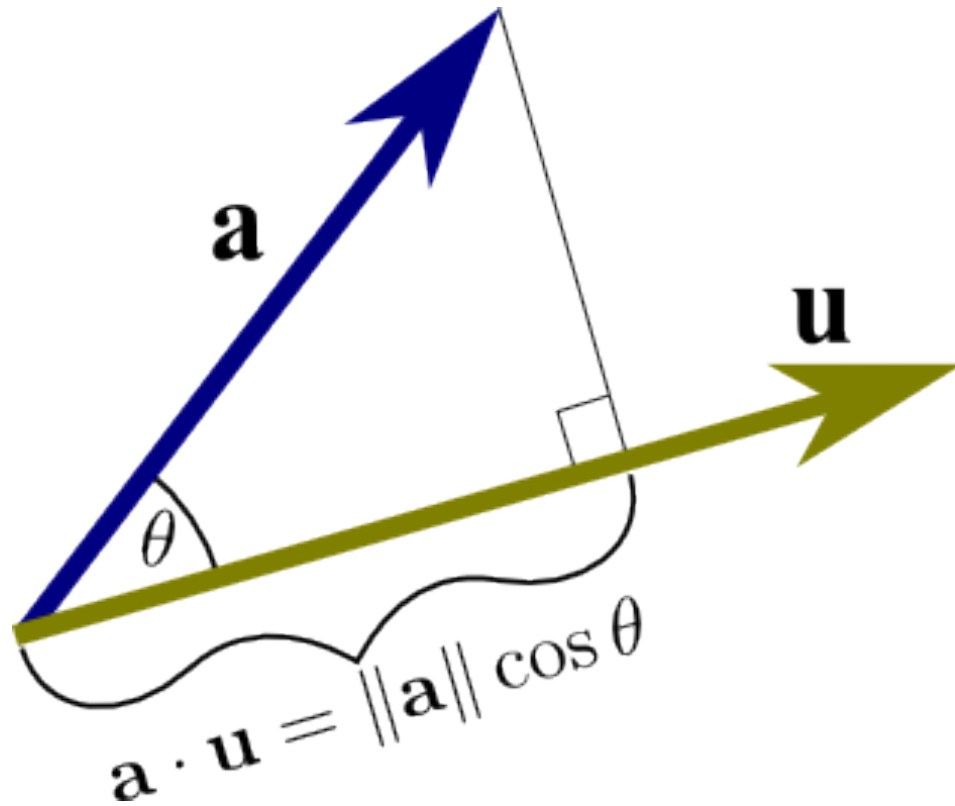
$$\vec{u} \cdot \vec{v} = (a\vec{i} + b\vec{j}) \cdot (c\vec{i} + d\vec{j}) = ac + bd$$

- Another formula for dot product:

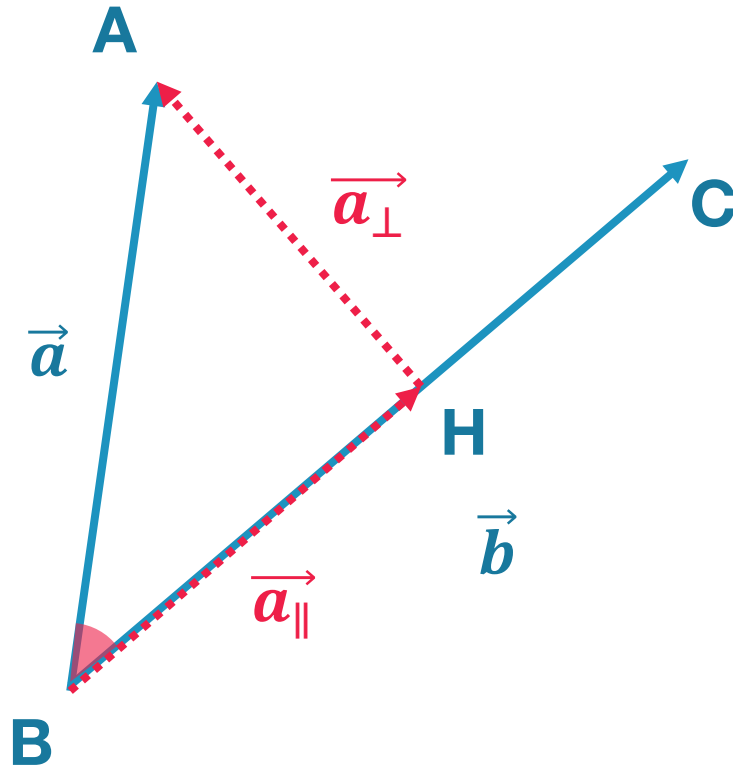
$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos(\vec{u}, \vec{v})$$

WELL, I KIND OF LIED...

- $\vec{u}$ is called the unit vector



VECTOR PROJECTION



- $\vec{a}_{\perp} = \vec{a} - \vec{a}_{\parallel}$
- $|\vec{a}_{\parallel}| = |\vec{a}| \cos \theta = |\vec{a}| \times \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$
 $|\vec{a}_{\parallel}| = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$
- $\vec{a}_{\parallel} = |\vec{a}_{\parallel}| \cdot \hat{b} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \hat{b}$
- $\hat{b} = \frac{1}{|\vec{b}|} \vec{b}$
- $\vec{a}_{\parallel} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$

- A triangle has vertices $A(\sqrt{3} + 1, -2, 4)$ $B(1, -2, 3)$ $C(2, -2, \sqrt{3} + 3)$. Find the area of the triangle.

- $$\begin{aligned} \text{Area} &= \frac{1}{2} AB \cdot BC \cdot \sin(\angle ABC) \\ &= \frac{1}{2} |\overrightarrow{AB}| |\overrightarrow{BC}| \cdot \sqrt{1 - (\cos \angle ABC)^2} \end{aligned}$$

- $$\begin{aligned} \overrightarrow{AB} &= (1 - (\sqrt{3} + 1))\vec{i} + (-2 - (-2))\vec{j} + (3 - 4)\vec{k} \\ &= -\sqrt{3}\vec{i} + 0\vec{j} - \vec{k} \end{aligned}$$

- $$\overrightarrow{BC} = \vec{i} + 0\vec{j} + \sqrt{3}\vec{k}$$

- A triangle has vertices $A(\sqrt{3} + 1, -2, 4)$ $B(1, -2, 3)$ $C(2, -2, \sqrt{3} + 3)$. Find the area of the triangle.

- $\cos \angle ABC = \frac{\vec{AB} \cdot \vec{BC}}{|\vec{AB}| |\vec{BC}|} = \frac{-2\sqrt{3}}{2 \times 2} = -\frac{\sqrt{3}}{2}$

- $$\begin{aligned} \text{Area} &= \frac{1}{2} |\vec{AB}| |\vec{BC}| \cdot \sqrt{1 - (\cos \angle ABC)^2} \\ &= \frac{1}{2} \times 2 \times 2 \times \sqrt{1 - \left(-\frac{\sqrt{3}}{2}\right)^2} \\ &= 2 \times \frac{1}{2} = 1 \end{aligned}$$

BREAK TIME!

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CLOSING REMARKS

- As the school year has been somewhat disrupted, this term is going to be a bit slower so that everyone can get readjusted.
- Make sure you keep your algebra game sharp! That means solving equations, factorization and graphing.
- Focus on Vectors, Complex numbers and Kinematics / Statics – these are the ones that will be followed up in Y12.
- Have fun and enjoy :D

EMPTY SLIDES IF TIME PERMITS
